

Integrated single-cell and spatial transcriptomics uncover distinct cellular subtypes involved in neural invasion in pancreatic cancer

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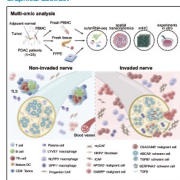
- 1 Literature Information
- 2 Research Background
- 3 Research Methods
- 4 Research Results
- 5 Discussion, Evaluation & Implication

Literature Information

Cancer Cell

Integrated single-cell and spatial transcriptomics uncover distinct cellular subtypes involved in neural invasion in pancreatic cancer

Graphical abstract



Highlights

- Multi-omics reveals TME differences in low- vs high-neural invasion PDAC tissues
- Non-invaded nerves link to TLS; NLRP3⁺ TAM and myCAF surround invaded nerves
- A distinct TGFBI⁺ Schwann cell subset promotes neural invasion
- Basal-like and GABRP⁺ malignant cells exhibit heightened neural invasion potential



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Article

Title

Integrated single-cell and spatial transcriptomics uncover distinct cellular subtypes involved in neural invasion in pancreatic cancer

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In brief

Chen et al. integrate sc/snRNA-seq with spatial transcriptomics to characterize cellular composition, lineage dynamics, and spatial architecture in PDAC tissues with varying neural invasion status, uncovering distinct peri- and intra-neural microenvironments and identifying TGFBI⁺ Schwann cells as promoters of neural invasion.

Study at a glance

- Human PDAC cohort with resected tumors.
- Integrated sc/snRNA-seq and spatial transcriptomics.
- Compare low- vs high-NI regions in the pancreas.
- Identify Schwann cells, stromal, immune and malignant subtypes associated with NI.

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Cancer as a Disease of the Microenvironment

- Traditional view: Cancer was considered mainly a disease of **malignant cells**.
- Modern view: Cancer is a complex ecosystem (Tumor microenvironment, **TME**) involving
 - Tumor cells
 - Stromal cells (fibroblasts, endothelial cells, etc.)
 - Immune cells
 - **Nerves and the nervous system**
- The nervous system regulates
 - Organ development
 - Tissue homeostasis
 - Immune function
- Increasing evidence: nerves actively participate in
 - Tumor growth and survival
 - **Invasion and metastasis**
 - Therapy resistance

Neural Involvement in Cancer

- Autonomic and sensory nerves can
 - Promote tumor proliferation and invasion
 - Shape the local immune microenvironment
 - Modulate responses to chemo- and radiotherapy
- Some studies also suggest **anti-tumor** roles of certain neural components.
- However, our knowledge is still limited:
 - Which specific neural and non-neural cell types are involved?
 - How do they interact within and around nerves?
 - How do these interactions drive disease progression?
- ⇒ Tumor-associated nerves represent a **critical but poorly understood** niche.

Neural (Perineural) Invasion

- **Neural invasion / perineural invasion (NI/PNI):**
 - Presence of **tumor cells within or around nerve structures**
 - Involving epineurium, perineurium, or endoneurium
- Common histopathological definitions:
 - Tumor cells identified **inside the nerve sheath**, or
 - Tumor cells surrounding a large proportion of the nerve circumference
- NI is a form of **local tumor spread**, distinct from:
 - Vascular invasion (via blood vessels)
 - Lymphatic invasion (via lymphatic vessels)
- Observed in many solid tumors:
 - **Pancreatic**, prostate, head and neck, colorectal, etc.

Basic Features and Clinical Impact of NI

- **Morphological features:**

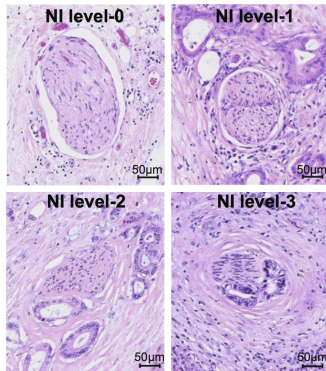
- Tumor cells tracking along nerves or clustering around nerve bundles
- Can involve small intrapancreatic nerves or larger extrapancreatic nerves

- **Biological implications:**

- Provides a **low-resistance path** for local tumor spread
- Allows tumor cells to reach distant sites along nerve routes
- Associated with nerve remodeling and inflammation

- **Clinical consequences:**

- Independent predictor of poor prognosis in several cancers
- Contributes to **local recurrence** and **distant metastasis**
- Strongly linked to **cancer-related neuropathic pain**



Neural Invasion in Pancreatic Ductal Adenocarcinoma (PDAC)

- The pancreas is a **highly innervated** organ.
- PDAC shows extremely high rates of NI/PNI:
 - Often reported in ~**80–100%** of resected tumors
- NI in PDAC is:
 - An **independent** adverse prognostic factor
 - Associated with **local recurrence** along nerve plexuses
 - Related to **distant metastasis** and **shortened survival**
- Clinically important symptoms:
 - Severe abdominal and back pain attributed to nerve involvement
- ⇒ Understanding NI in PDAC is crucial for
 - Better risk stratification and prognosis
 - Development of **nerve-targeted** therapeutic strategies

Mechanistic View: How Does NI Occur?

- **Bidirectional crosstalk** between tumor cells and nerves:
 - Tumor cells secrete neurotrophic factors and chemokines
 - Nerves release growth factors and neurotransmitters
- Stromal and immune cells in perineural regions:
 - Support tumor cell survival and migration
 - Modulate local inflammation and pain
- Conceptual model:
 - Tumor cells “**hijack**” **neural pathways** as routes of spread
 - Nerves are reshaped into a **pro-tumor niche**
- **Knowledge gap**:
 - The exact **cellular composition** of this nerve-associated niche
 - How different cell subtypes collectively drive NI in PDAC
 - ⇒ Motivates single-cell and spatial dissection in this study

Limitations of Existing Studies on Neural Invasion

- Previous work has focused on
 - Neurotrophic factors, chemokines, axon guidance molecules
 - *In vitro* co-culture systems and animal models
- These studies highlight **bidirectional signaling** between tumor cells and nerves.
- Stromal and immune cells are also implicated in neural invasion.
- However:
 - Most studies examine **single molecules** or **single cell types**.
 - There is no comprehensive map of the **cellular ecosystem** inside and around nerves.
 - Translation of these findings to clinical practice remains limited.

Advances in Single-Cell and Spatial Omics

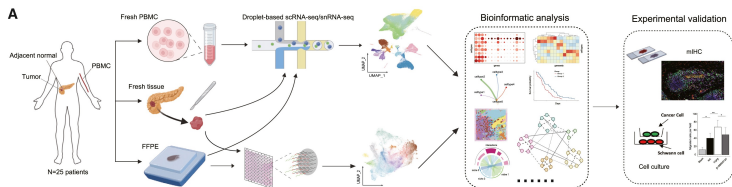
- Single-cell/single-nucleus RNA sequencing has
 - Revealed diverse malignant cell states in PDAC
 - Characterized stromal and immune cell heterogeneity
- Spatial transcriptomics has
 - Linked cell states to their anatomical locations
 - Uncovered spatially organized tumor-stroma niches
- Despite these advances:
 - Most studies focus on **bulk tumor and stromal regions**
 - The **nerve-associated compartment** (inside and around nerves) remains poorly characterized
- ⇒ A systematic, spatially resolved map of nerve-associated microenvironments in PDAC is still missing.

Rationale of This Study

- **Biological and clinical need:**
 - Neural invasion is highly prevalent and clinically important in PDAC.
 - The cellular and molecular mechanisms within nerves and perineural regions are not well defined.
- **Technological opportunity:**
 - Single-cell/single-nucleus RNA-seq can resolve cell types and states at high resolution.
 - Spatial transcriptomics can preserve anatomical context, including nerve structures.
- **Overall goal of the paper:**
 - To integrate single-cell/single-nucleus and spatial transcriptomic data from PDAC.
 - To delineate **distinct cellular subtypes** within nerve-associated microenvironments.
 - To uncover how these subtypes contribute to **neural invasion** in pancreatic cancer.

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Research Methods Overview



- Multi-region PDAC cohort integrating:
 - Single-cell / single-nucleus RNA sequencing (sc/snRNA-seq)
 - Spatial transcriptomics (Visium on fresh-frozen and FFPE)
 - Histology & multiplex immunohistochemistry / immunofluorescence
 - *In vitro* functional assays
- Focus: cellular ecosystem and neural invasion (NI) in human PDAC
- Strategy:
 - Build single-cell atlas of tumors, adjacent pancreas, and blood
 - Map cell states in space using ST
 - Model cell-cell communication and validate key interactions

Patient Cohort & Tissue Sampling

- Fresh surgical PDAC specimens
 - Primary tumor regions
 - Adjacent non-tumor pancreas
 - Peripheral blood (PBMCs)
 - In total 62 samples from 25 PDAC patients
- Standard histology
 - H&E staining of representative blocks
 - Pathologist review of tumor, stroma, nerves
- Neural invasion (NI) assessment
 - Identification of perineural invasion (PNI)
 - Semi-quantitative NI score per case
 - Classification into low-NI (8 patients) vs high-NI (17 patients) tumors

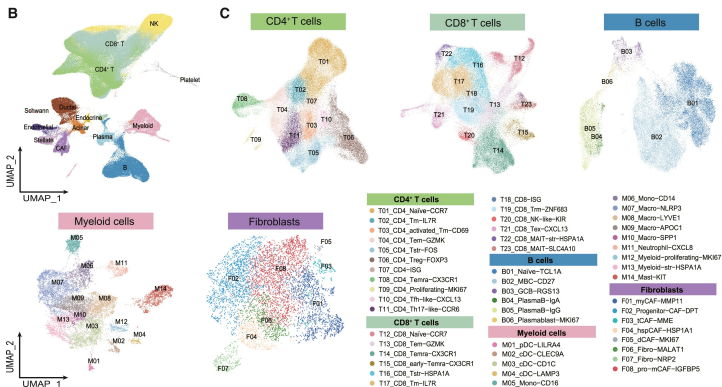
sc/snRNA-seq Data Generation

- Tissue dissociation and nuclei isolation
 - Fresh tumor: single-cell suspensions for scRNA-seq
 - Frozen / less viable tissue: nuclei for snRNA-seq
- Library preparation & sequencing
 - 10x Genomics platform (scRNA-seq and snRNA-seq)
 - Generation of gene-cell count matrices
- Quality control
 - Filtering low-complexity cells/nuclei
 - Removing cells with high mitochondrial gene percentage
 - Doublet and ambient RNA removal (standard pipelines)

Single-Cell Data Processing & Annotation

- Data integration
 - Batch correction across patients, tissues, sc vs sn
 - Dimensionality reduction (PCA, UMAP)
 - Unsupervised clustering
- Major lineage annotation
 - Canonical markers for T/NK, B/plasma, myeloid, CAF/fibroblast
 - Endothelial, stellate, acinar, endocrine, ductal, Schwann cells
- Malignant vs non-malignant ductal cells
 - Inferred large-scale CNVs from expression profiles
 - Definition of malignant ductal subtypes
- Subclustering
 - Finer states within tumor, immune, stromal, and Schwann cells

Single-Cell Data Processing & Annotation

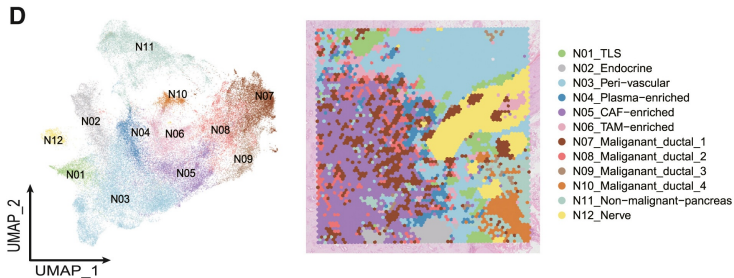


- B: Data integration and major lineage annotation
- C: Subclustering within each major lineage

Spatial Transcriptomics

- Platforms & samples
 - 10x Visium on fresh-frozen tissue sections
 - CytAssist-enabled Visium on FFPE sections
 - Adjacent H&E sections for morphological annotation
- ST data processing
 - Alignment to reference genome, spot-level count matrices
 - QC, normalization, dimensionality reduction, clustering
- Cell-type deconvolution
 - Projection of sc/snRNA-seq signatures onto ST spots
 - Identification of spatial “niches” enriched for specific malignant, immune, CAF, and Schwann subtypes

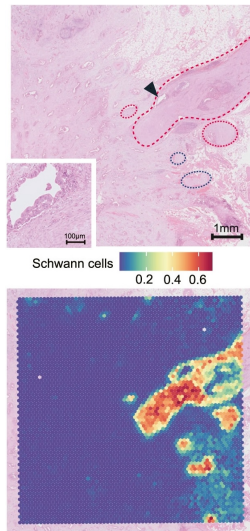
Spatial Transcriptomics



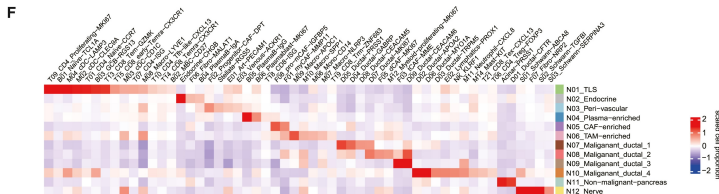
- UMAP plots showing spatial niches clustering based on gene expression of the spots (left). Mapping of spatial niches in a representative sample PA#98 (right).

NI-Focused Spatial Analyses

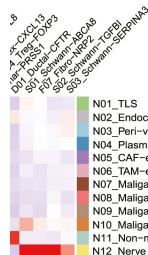
- Nerve annotation
 - Pathologists labeled nerve bundles on H&E (Top)
 - Classification: invaded vs non-invaded nerves
 - Black arrow indicates a magnified field displaying invading cancer cells
 - Deconvolution-based Schwann cell abundance around nerves (Bottom)
- Distance-based analysis
 - ST spots grouped by distance to nerves (juxta-neural rings, peri-neural regions)
 - Comparison of cell-type composition and gene signatures around invaded vs non-invaded nerves



NI-Focused Spatial Analyses



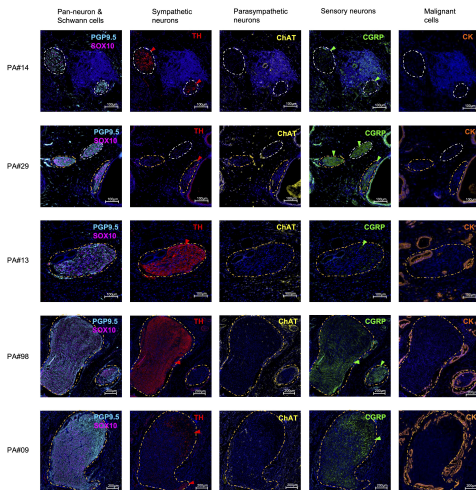
- Integration with tumor regions
 - Mapping of tumor, stromal, and TLS-like niches
 - Analysis of nerve-tumor interfaces and associated cell states
 - Heatmap of cell proportions per spatial niche (top)
 - Schwann cells predominantly located in the N12_Nerve niche (right)



Cell-Cell Communication & Multi-Omics Integration

- Ligand-receptor inference (sc/snRNA-seq)
 - Modeling signaling between malignant cells, CAFs, myeloid cells, T cells, and Schwann subsets
 - Focus on pathways related to neural remodeling and immunity
- Cytokine signaling into Schwann cells
 - Estimation of pathway activities (e.g. TGF- β)
 - Association with distinct Schwann cell states
- Spatial interaction modelling
 - Construction of spatial networks from ST data
 - Quantifying co-localization and influence across niches

Imaging Validation & Functional Assays



- Multiplex IHC / IF
- Gold dashed lines indicate invaded nerves, while white dashed lines indicate non-invaded nerves
- Nerve bundles comprised both TH⁺ sympathetic and CGRP⁺ sensory neurons, whereas ChAT⁺ parasympathetic neurons were relatively rare

- Both sympathetic-dominant nerves and sensory-dominant nerves were found to be susceptible to invasion

Imaging Validation & Functional Assays

- *In vitro* Schwann cell assays
 - Human Schwann cell line treated with $\text{TGF-}\beta \pm \text{TGF-}\beta$ receptor inhibitor
 - Transcriptomic comparison with *in vivo* Schwann states
- Tumor cell migration assays
 - PDAC cells exposed to different Schwann-conditioned media
 - Time-lapse imaging to measure migration velocity

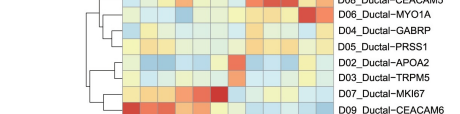
Statistical Analysis

- Group comparisons
 - Low-NI vs high-NI tumors
 - Invaded vs non-invaded nerves
 - Tests: mainly two-sided Wilcoxon rank-sum tests
- Correlation analyses
 - NI scores vs clinical/pathological variables
 - Pearson correlation with confidence intervals
- Enrichment & pathway analysis
 - Gene set enrichment for pathways (e.g. EMT, neural signaling)
 - Diversity indices (e.g. Shannon index) for malignant subtypes

- ## 5 Discussion, Evaluation & Implication

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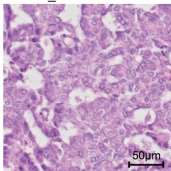
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- D08_Ductal-CEACAM5: “classical” subtype



- D08_Ductal-CEACAM5:
“classical” subtype.
- D09_Ductal-CEACAM6:
“basal-like” subtype.
- D04_Ductal-GABRP,
D05_Ductal-PRSS1: intermediate
(mixed signatures).
- D02_Ductal-APOA2:
neuroendocrine-like program.

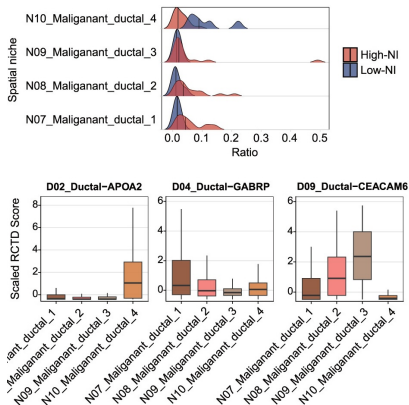
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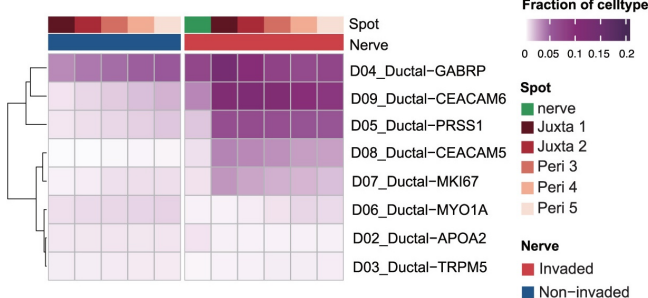
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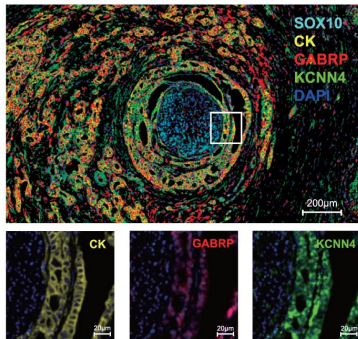
- Low-NI group:
 - Enrichment of D02_Ductal-APOA2 (neuroendocrine-like).
 - Malignant_ductal_4 niche (rich in D02) more prevalent.
 - Lower NI-related gene scores in these cells.





- D09_Ductal-CEACAM6: pro-migratory and pro-invasive features
 - Enriched pathways: **integrin-mediated signaling, ameboid-type migration, adhesion-dependent spreading**
 - Exhibited the **highest EMT score** among malignant subtypes
 - **WNT signaling** elevated, consistent with EMT activation
 - Upregulated **CTSA** and **CTSB** (ECM-degrading proteinases)
- Interpretation
 - D09_Ductal-CEACAM6 likely has a **heightened capability of migrating through the tissue matrix**,
 - Spatial enrichment near invaded nerves suggests it may **actively drive neural invasion**

- D04_Ductal-GABRP marker gene: GABRP
 - GABRP encodes a receptor subunit for the inhibitory neurotransmitter **GABA**
 - D04_Ductal-GABRP is characterized by **high GABRP expression**
- High nerve-interaction potential
 - A curated NI-related, nerve-associated gene set was applied
 - **D04_Ductal-GABRP scored the highest** among all malignant subpopulations
 - ⇒ Suggests **active reciprocal interactions with nerves** that may promote NI
- Not dependent on GABAergic innervation
 - GAD65 used as a marker for GABAergic neurons in mIHC
 - **No GAD65⁺ neurons** detected, even in tumors rich in GABRP⁺ malignant cells
 - Consistent with reports that **GABRP promotes tumor migration in a GABA-independent, KCNN4-synergistic manner**
- GABRP-KCNN4 axis in high-NI tissues
 - mIHC confirmed **co-expression of GABRP and KCNN4** in malignant cells in high-NI tissues



- mIHC staining showing absence of GAD65+ GABAergic neurons in PDAC tissues with high proportion of GABRP+ malignant cells
- mIHC staining showing co-expression of GABRP and KCNN4 in malignant cells

- **0.1 decimal**

- 9 ductal subclusters (D01–D09) identified by sc/snRNA-seq.
- D01_Ductal-CFTR: low CNV, transcriptionally normal ductal cells.
- D02–D09: malignant ductal states spanning **classical**, **basal-like**, intermediate and **neuroendocrine-like** programs.

- **Four's rule** – 1 factor

- Enriched for D02_Ductal-APOA2 (neuroendocrine-like, Malignant_ductal_4 niche).
- High lipid/cholesterol metabolism, low NI-related gene scores and low EMT score ⇒ **less invasive phenotype**.

- Reduced mal

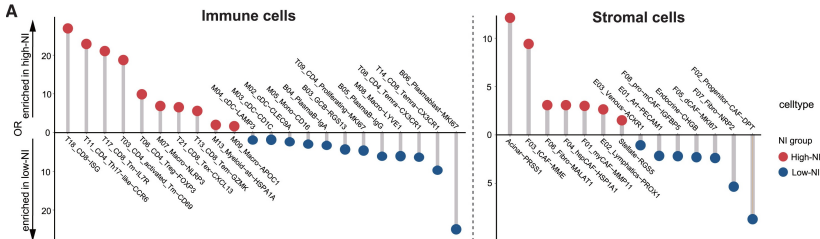
- Reduced malignant subtype diversity; dominated by basal-like, intermediate and classical states.
- D09_Ductal-CEACAM6 (basal-like): strong EMT, integrin-mediated migration, WNT signaling and ECM proteases $\Rightarrow$ **high intrinsic migratory/invasive capacity**.
- D04_Ductal-GABRP (intermediate): highest NI-related gene-set scores and enriched near invaded nerves; GABRP-KCNN4 axis supports tumor migration in a **GABA-independent** manner.

- Question: How do immune & stromal compartments differ between low- and high-NI PDAC?
- Strategy:
 - Integrate sc/snRNA-seq and ST from 21 PDAC patients (5 low-NI, 16 high-NI).
 - Quantify enrichment using ratio of observed to expected cell numbers (Ro/e), miloR and spatial deconvolution.
 - Zoom into juxta-neural regions of invaded vs non-invaded nerves.
- Key message:
 - Low-NI tumors are enriched for TLS-centered immunity, whereas high-NI tumors show inflammatory and immunosuppressive microenvironments around nerves.

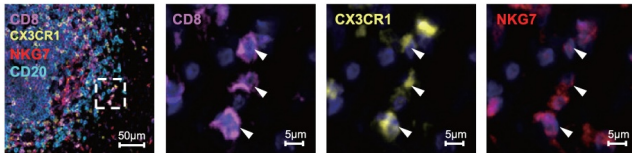
Low-NI TME: TLS-centered anti-tumor immunity

- Evidence for **increased TLS abundance in low-NI**
 - GCBs primarily reside in **TLSs** and differentiate into plasma cells within TLSs
 - ST showed higher proportions of key TLS components in low-NI tissues:
 - B01_naive-TCL1A
 - B03_GCB-RGS13
 - T10_CD4_Tfh-like CXCL13
- **TLS-/B cell-related spatial niches in low-NI**
 - PCA of spatial niches revealed a **distinct cluster** for low-NI tissues
 - Low-NI group showed higher proportions of:
 - **N01_TLS niche**
 - N04_Plasma-enriched niche
 - N03_Perivascular niche

⇒ Organized **TLS-like structures** and **B cell immunity** are associated with the **low-NI status** in PDAC



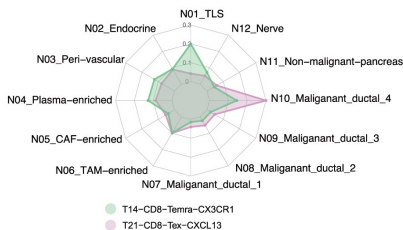
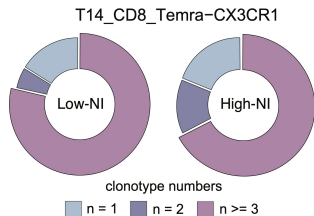
- Clonal expansion and migration of CD8₊ Temra in low-NI
 - scTCR-seq + STARTRAC used to trace T cell clonal dynamics
 - In low-NI group:
 - CD8₊ Temra showed **robust clonal expansion** in tumors
 - Displayed the **highest migration from blood** among T cell subsets
 - ⇒ Origin from both **in situ expansion** and **peripheral blood recruitment**
- Spatial localization: TLS-/plasma-/vascular niches
 - ST localized CD8₊ Temra mainly to N01, N04 and N03 niche.
 - Consistent with **expansion within TLS-like structures** and **migration from vasculature**
- Histology: CD8₊ Temra observed at the **periphery of TLS-like structures**, mirroring plasma cell distribution



CD8 T cell dynamics and TLS in low-NI vs high-NI PDAC

- High-NI: **weakened effector CD8 response, more exhaustion**
 - In high-NI group, CD8_Temra showed **reduced expansion and migration** and lower representation in N01_TLS niche
 - **Tex (exhausted CD8 T cells)** were more abundant in high-NI tissues
 - Tex preferentially localized to **ductal-enriched niches**, not TLS-related regions

⇒ TLSs may generate effector CD8 T cells counteracting invasion in low-NI, whereas high-NI tumors exhibit **exhausted, TLS-independent CD8 immunity**

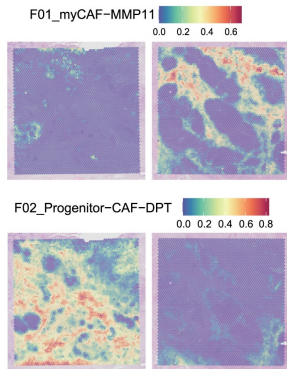


High-NI TME: inflammatory and immunosuppressive

- **Regulatory T cells:**
 - T06_CD4_Treg-FOXP3 proportion increased in high-NI tumors.
 - Likely contributes to **suppression of anti-tumor immunity**.
- **Pro-inflammatory macrophages:**
 - M07_Macro-NLRP3 enriched in high-NI tissues.
 - High IL1B expression and pathways related to **inflammatory response**.
- Th17-like CD4⁺ T cells:
 - Co-occurrence of M07_Macro-NLRP3 with T11_CD4_Th17-like-CCR6, suggesting TAM-driven Th17 polarization.
- Implication: high-NI TME combines **chronic inflammation with immunosuppression**, promoting tumors.

Stromal CAF subtypes reprogramming in high-NI tumors

- **myCAF and tCAF expansion:**
 - F01_myCAF-MMP11 and F03_tCAF-MME increased in high-NI tissues.
 - Express **ECM remodeling, hypoxia and angiogenesis-related genes.**
- **Loss of progenitor-like CAFs:**
 - F02_Progenitor-CAF-DPT (PI16⁺ progenitor-like CAFs) relatively enriched in low-NI tissues.
- **ST validation:**
 - External ST cohort confirms myCAF enrichment in high-NI and progenitor-CAF predominance in low-NI.
- **Implication: stromal remodeling in high-NI tumors may facilitate nerve invasion and tumor spread.**



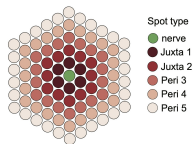
- Low NI (left), high NI (right)

Part 2 Summary: TME states linked to NI status

- **Low-NI tumors: TLS-centered, effector-dominant immunity**
 - Enriched in **germinal center B cells**, plasmablasts and plasma cells.
 - Higher abundance of **TLS components** and TLS-/plasma-/perivascular niches (N01, N04, N03).
 - Clonally expanded, cytotoxic **CD8₊ Temra CX3CR1** cells localize to TLS/perivascular regions.
- **High-NI tumors: inflammatory and immunosuppressive niches**
 - More **exhausted CD8₊ Tex** in ductal-enriched niches; reduced CD8₊ Temra expansion and migration.
 - Increased **Tregs**, **NLRP3⁺ macrophages** and Th17-like CD4⁺ T cells.
 - Expansion of **myCAF/tCAF** and loss of progenitor-like CAFs; myCAF/tCAF accumulate around invaded nerves.
- **Conceptual model**
 - **Low-NI:** “protective” TLS-rich, B/Temra-supported microenvironments around nerves.
 - **High-NI:** “pro-invasive” niches dominated by inflammation, immunosuppression and CAF-driven remodeling.

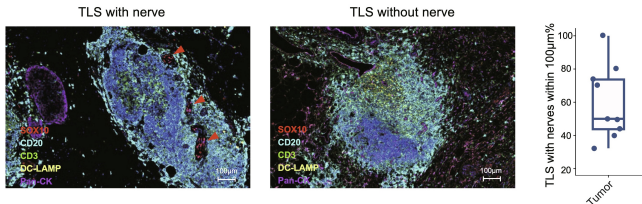
Part 3 Overview

- Question:
 - How do the cellular and molecular landscapes differ around **non-invaded** vs **invaded** nerves?
- Strategy:
 - Define “nerve spots” (overlapping with nerves) and juxta-neural rings (two layers, $\sim 200\ \mu\text{m}$) around nerves in ST data.
 - Compare cell-type composition, spatial niches and signaling pathways between non-invaded and invaded nerves.
 - Integrate sc/snRNA-seq, ST, COMMOT communication analysis, MISTy spatial modelling and mIHC.
- Key message:
 - **Non-invaded nerves** are embedded in **TLS-rich, adaptive immune** microenvironments.
 - **Invaded nerves** are surrounded by **innate inflammatory** and **T cell-exhausted** niches with stromal remodeling.



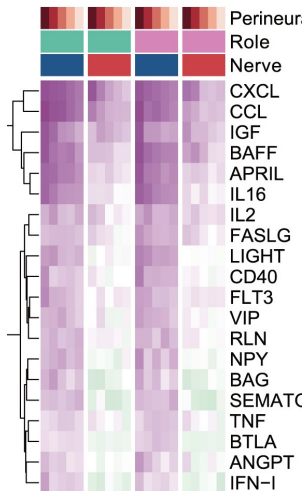
Nerve-TLS structures (NTSs) around non-invaded nerves

- **Cellular composition** around non-invaded nerves
 - Juxta-neural regions enriched for **TLS components**: B03_GCB-RGS13 (germinal center B cells), T10_CD4.Tfh-like CXCL13, M04_cDC-LAMP3.
 - Fractions of N01_TLS and N04_Plasma-enriched niches **increase** as distance to non-invaded nerves declines.
 - This spatial gradient is **not** observed around invaded nerves.
- **Nerve-TLS association**
 - Among 141 non-invaded nerve fibers in tumors, $\sim 64\%$ had a TLS within $100\ \mu\text{m}$ of the nerve border.
 - Non-invaded nerves in adjacent normal tissues were **less** associated with TLSs $\Rightarrow$ TLS formation around nerves is **tumor-associated**.
 - Among all tumor TLSs, $\sim 50\%$ had a nerve fiber within $100\ \mu\text{m}$ $\Rightarrow$ **nerve-TLS structures (NTSs) are common in PDAC**.

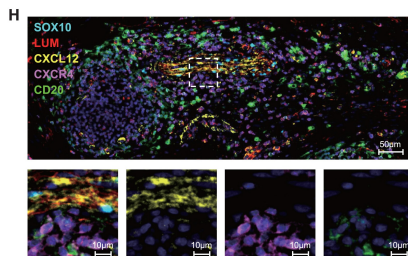
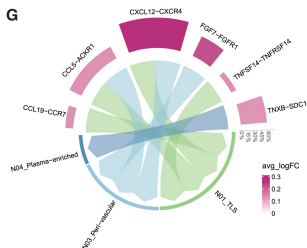


Signaling environment around non-invaded vs invaded nerves

- **COMMOT analysis** of juxta-neural neighborhoods
 - Optimal-transport-based inference of spatial ligand-receptor signaling between cell populations
 - Modelled secreted signaling, cell-cell contact and ECM-receptor pathways using ligand-receptor expression
- Non-invaded nerves: **adaptive immune-supportive signals**
 - Elevated pathways for **leukocyte recruitment**: CXCL, CCL, IL-16.
 - Elevated pathways for **lymphocyte proliferation and differentiation**: BAFF, APRIL, IL-2, LIGHT, CD40.
 - Indicates an **active signaling environment** supporting TLS construction and adaptive immune responses.



Signaling environment around non-invaded vs invaded nerves

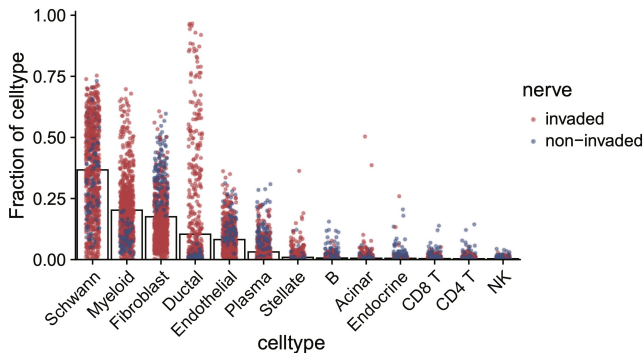


- **CXCL12-CXCR4 axis and TLS organization**
 - Comparing non-invaded vs invaded nerve spots highlighted candidate ligand-receptor pairs.
 - mIHC showed abundant CXCL12 in fibroblasts inside and around nerves.
 - Suggests that **nerve-associated fibroblasts** may secrete CXCL12 to recruit CXCR4⁺ B cells and facilitate TLS assembly.
 - Aligns with previous reports implicating stromal- and epithelial-derived CXCL12 in TLS induction.

Invaded nerves: inflammatory and immunosuppressive niches

- Immunosuppressive pathways around invaded nerves
 - Increased **IL-10 and TGF- β signaling** in juxta-neural regions.
 - Higher expression of **T cell exhaustion markers** than around non-invaded nerves.
- **Pro-inflammatory IL-1 signaling**
 - Upregulated IL-1 pathway around invaded nerves.
 - M07_Macro-NLRP3, enriched near invaded nerves, expresses high IL1B, reminiscent of **wound-like** inflammatory responses.
- Conceptual contrast
 - **Non-invaded nerves:** TLS construction and adaptive immune protective environment.
 - **Invaded nerves:** innate inflammatory niche (inflammatory macrophages and neutrophils) combined with **T cell exhaustion** and immunosuppression.

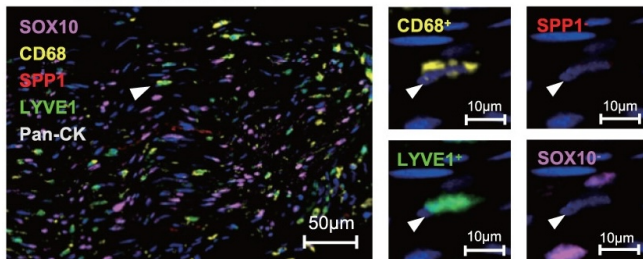
Cellular composition of non-invaded and invaded nerves

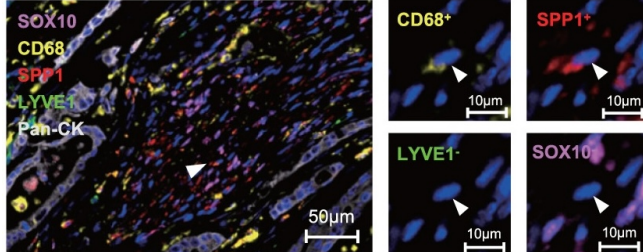


- Nerve-resident stromal composition
 - ST analysis determined that within tumor-associated nerves, **Schwann cells**, **myeloid cells** and **fibroblasts** are the three most abundant cell types.

Endoneurial fibroblasts: F07_Fibro-NRP2

- Identification of a nerve-specific fibroblast subset
 - Unsupervised clustering of sc/snRNA-seq revealed **F07_Fibro-NRP2** as a fibroblast subset separated from other fibroblasts on UMAP.
 - Shares LUM and DCN expression with other fibroblasts, but distinctly **upregulates axonogenesis-related genes**: NGFR, NRP2, AUTS2, RGMA.
- Spatial localization and markers
 - Projection onto ST data showed F07_Fibro-NRP2 specifically located **inside nerves**.
 - MISTy modelling revealed **strong spatial vicinity to Schwann cells** $\Rightarrow$ a distinct **endoneurial fibroblast** subset.
 - Endoneurial fibroblasts specially express APOD (lipoprotein), yielding APOD⁺LUM⁺ fibroblasts in nerves, confirmed by mIHC.
- Functional inference
 - Pathway analysis suggests roles in **axonogenesis, regulation of stem cell differentiation, vasculogenesis and myeloid cell homeostasis**.

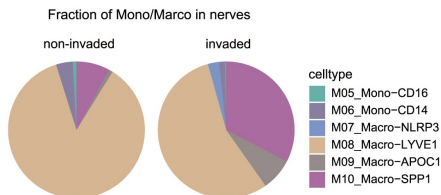
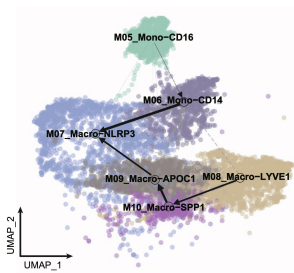




Nerve-resident macrophages: from LYVE1⁺ to SPP1⁺

• Trajectory and interpretation

- RNA velocity and PAGA suggest **M10_Macro-SPP1** may arise from **M08_Macro-LYVE1**.
- Monocyte infiltration into nerves is limited $\Rightarrow$ nerve-resident macrophages likely **convert into repair-like SPP1⁺ macrophages** in response to cancer invasion.
- Further experiments are needed to validate these developmental trajectories and functions.



Part 3 Summary: Juxta-neural niches and nerve-resident cells

- **Juxta-neural niches: protective TLS vs inflammatory niches**
 - **Non-invaded nerves:** Juxta-neural regions enriched for **TLS components**; Frequent NTSs suggest a **tumor-associated, adaptive immune shield** supported by CXCL12-CXCR4 signaling.
 - **Invaded nerves:** Juxta-neural regions show **increased IL-10 and TGF- β signaling**, higher **T cell exhaustion** marker expression, and **IL-1-driven innate inflammation**, forming a wound-like pro-invasive niche.
- **Nerve-resident stromal and myeloid cells**
 - Identified **F07.Fibro-NRP2** as **APOD⁺LUM⁺** endoneurial fibroblasts enriched in axonogenesis, vasculogenesis and myeloid homeostasis pathways.
 - Nerves harbor **LYVE1⁺ tissue-resident macrophages** (M08) in non-invaded states and **SPP1⁺TREM2⁺ macrophages** (M10) in invaded states.
 - Trajectory analysis suggests that **resident macrophages may convert into SPP1⁺ repair-like macrophages** upon cancer invasion, rather than being derived from infiltrating monocytes.
- **Overall**
 - Juxta-neural niches transition from **TLS-rich adaptive protection to innate inflammatory, immunosuppressive and stromally remodeled environments** as nerves become invaded in PDAC.

Part 4 Overview

- Question: how do Schwann cell states contribute to neural invasion?
- Approach:
 - Define Schwann subclusters from sc/snRNA-seq.
 - Map their spatial distribution relative to tumor and nerves.
 - Use computational ligand-receptor analysis and *in vitro* assays to infer upstream signals and functional roles.
- Key message:
 - A distinct TGFBI⁺ Schwann subset (S02) is induced by TGF- β 1 and promotes cancer cell migration and neural invasion.

Schwann cell heterogeneity in PDAC

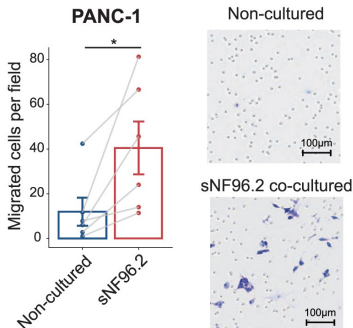
- Unsupervised clustering of Schwann cells identified **three Schwann subsets**.
- **S01_Schwann-ABCA8: myelinating Schwann cells**
 - High expression of myelin-related genes: **MPZ, MBP, EGR2**.
 - ABCA8 involved in **myelin synthesis** $\Rightarrow$ **myelinating Schwann cell phenotype**.
- **S02_Schwann-TGFBI: ECM- and growth factor-rich subset**
 - Lower myelin gene expression but higher **ECM genes**: **FN1, COL12A1, CCN3, TGFBI**.
 - TGFBI can be induced by **TGF- β** signaling.
 - Upregulated growth factors: **FGF5, MDK, BTC** $\Rightarrow$ suggests a potential **pro-tumorigenic role**.
- **S03_Schwann-SERPINA3: repair-like Schwann cells**
 - Express AP-1 family genes: **JUNB, JUND, FOS**, resembling **repair Schwann cells** in Bungner's bands (Schwann cell tubes that guide regenerating axons after peripheral nerve injury).
 - Pathways enriched in **stress response**, regulation of endopeptidase activity and fibroblast proliferation $\Rightarrow$ responds to injury and initiates repair signals.

-
- SOX10
TGFBI
MPZ
Pan-CK
- 50 μm

Functional properties of S02_Schwann-TGFBI

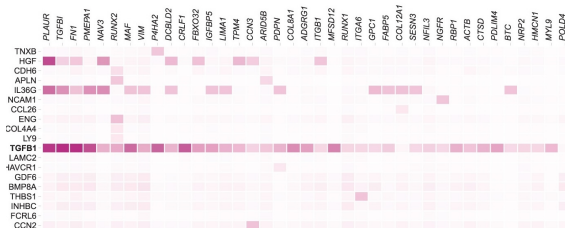
- ECM remodeling and growth factor production
 - Compared with S01 and S03, S02_Schwann-TGFBI expresses higher levels of genes **constituting or regulating the ECM**, which can facilitate cell migration and metastasis.
 - Upregulates multiple **growth factor-encoding genes**, potentially supporting cancer cell **survival and proliferation**.
- Pathway analysis indicated enhanced activities in:
 - Cadherin binding, focal adhesion, integrin-mediated signaling ⇒ **strong adhesion and migration properties**.
 - Regulation of neuron projection development, neuron migration, neuron-to-neuron synapse ⇒ **potential to guide neuron projections**.
- Overall, S02_Schwann-TGFBI may **create a favorable microenvironment** for cancer cell migration and invasion by ECM remodeling, matrix adhesion and neuro-guidance.

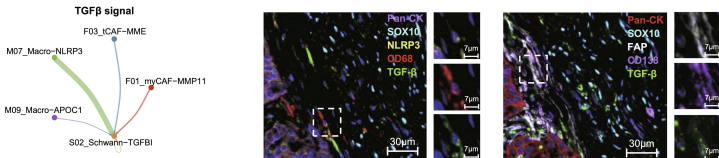
- Schwann cell line model:
 - RNA-seq demonstrates that sNF96.2 transcriptomic profiles **closely match S02_Schwann-TGFBI** signatures, rather than S01 or S03.
- ⇒ Supports that **S02-like Schwann cells actively promote cancer cell migration**.



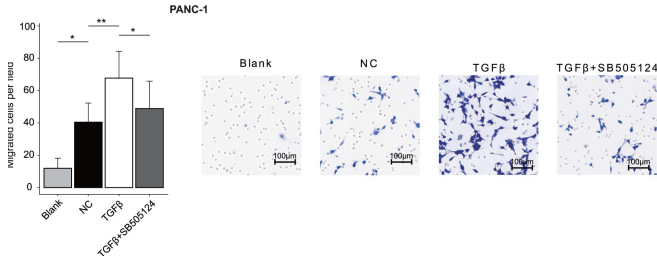
- Regulon analysis: TCF β linked transcription factors

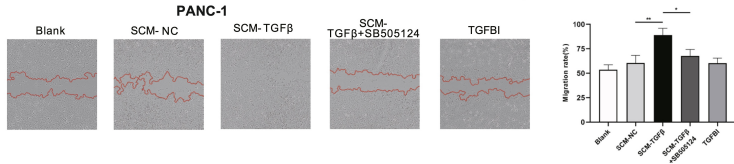
- **Regulon analysis: TGF- β -linked transcription factors**
 - SCENIC-based regulon analysis on S02_Schwann-TGFBI revealed distinct activity of **RUNX1**, **TFAP2A** and **SMAD1**.
 - These transcription factors are known to participate in **TGF- β signaling**, suggesting that S02_Schwann-TGFBI is shaped by microenvironmental TGF- β cues.
- **Candidate inducing cytokines**
 - CytoSig and NicheNet were used to predict upstream ligands regulating the S02 signature.
 - **TGF- β 1** and **HGF** ranked among the top candidate factors.





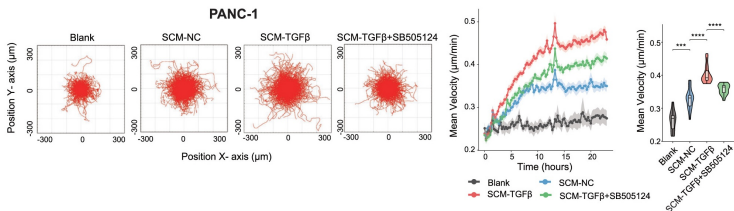
- **TGF- β 1 sources around invaded nerves**
 - Four subsets enriched in high-NI tumors and adjacent to invaded nerves: **M07_Macro-NLRP3**, **M09_Macro-APOC1**, **F01_myCAF-MMP11**, **F03_tCAF-MME**.
 - All can potentially produce TGF- β 1, with **M07** and **F01** as major contributors (higher expression and/or greater cell proportions).
 - CellChat analysis shows strong **TGF- β signaling from M07 and F01 to S02_Schwann-TGFBI**.
 - mIHC confirms TGF- β 1 expression by **NLRP3⁺ TAMs** and **myCAFs**.
- **HGF less likely to be dominant**
 - HGF is barely expressed by cells adjacent to invaded nerves, arguing against a major role in S02 induction.
- $\Rightarrow$ **S02_Schwann-TGFBI is likely induced by TGF- β 1 produced from M07_Macro-NLRP3 and F01_myCAF-MMP11 around invaded nerves.**





In vitro validation of TGF- β 1 induction of TGFBI⁺ Schwann cells

- TGF- β 1-treated Schwann cells promote PDAC cell migration & invasion
 - **Random migration assays:**
 - Conditioned medium from TGF- β 1-treated sNF96.2 increases **velocity** and **accumulated distance** of PANC-1 and MIA PaCa-2 cells.
- TGFBI protein shows **minimal effect** in these assays.
- $\Rightarrow$ TGF- β 1-induced Schwann cell state (not TGFBI alone) drives pro-migratory effects.



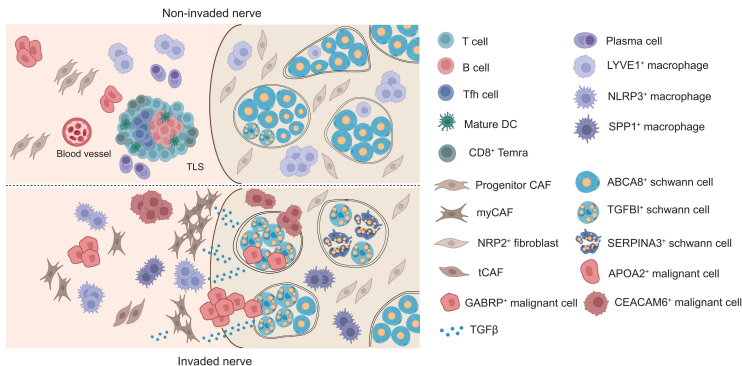
Part 4 Summary: TGFBI⁺ Schwann subset induced by TGF- β 1 promotes neural invasion

- Schwann cell states:
 - S01_Schwann-ABCA8: myelinating, homeostatic.
 - S02_Schwann-TGFBI: TGF- β -driven transforming state with ECM-remodeling and pro-invasive features.
 - S03_Schwann-SERPINA3: stress/inflammation-associated.
- High-NI context:
 - Pro-inflammatory macrophages (M07) and myCAFs (F01) around invaded nerves secrete TGF- β 1.
 - TGF- β 1 induces S02_Schwann-TGFBI, which in turn promotes cancer cell migration and nerve invasion.
- Integrated model of NI in PDAC:
 - Intrinsic: basal-like CEACAM6⁺ and GABRP⁺ malignant cells have high NI potential.
 - Microenvironmental: loss of TLS-rich protective niches and gain of inflammatory, CAF- and TGFBI⁺ Schwann cell-rich niches drive neural invasion.

- ## 5 Discussion, Evaluation & Implication

Discussion I: Key Findings

- Built an integrated single-cell and spatial atlas of PDAC with different neural invasion (NI) statuses, resolving the cellular composition and organization around nerves.



Discussion I: Key Findings

- **Identified a novel intraneural fibroblast subset (F07_Fibro-NRP2):**
 - Enriched for axonogenesis- and neurogenesis-related genes.
 - Preferentially localized inside nerves.
 - High APOD expression suggests a role in mitigating oxidative stress and supporting neuronal integrity.
- **Revealed marked heterogeneity of Schwann cells:**
 - S01_Schwann-ABCA8: myelinating-like Schwann cells.
 - S03_Schwann-SERPINA3: repair-like Schwann cells.
 - S02_Schwann-TGFBI: distinct, ECM- and growth factor-rich subset associated with NI.
- External datasets and *in vitro* assays support that **TGF- β 1 drives the induction of S02-like TGFBI⁺ Schwann cells and enhances PDAC cell migration and invasion.**

Discussion II: Mechanistic Model and Implications

- At the invasive neural front, **myCAF and NLRP3⁺ TAMs are enriched, while LYVE1⁺ tissue-resident macrophages and progenitor-like CAFs are depleted.**
- **MyCAF and NLRP3⁺ TAMs produce high levels of TGF- β 1:**
 - Strengthened TGF- β signaling around invaded nerves.
 - Promotes reprogramming of Schwann cells toward the S02_Schwann-TGFBI phenotype.
- **Identified NI-associated malignant ductal subsets:**
 - D09_Ductal-CEACAM6 resembles previously reported “transition/basal-like” high-NI states.
 - D04_Ductal-GABRP shows a nerve-responsive transcriptional program, indicating multiple routes to acquire neuro-invasive potential.
- **Broader implications:**
 - Provides a human PDAC nerve-centric atlas to guide future targeting of specific stromal and immune subsets at the NI front.
 - Offers a framework for dissecting neuro-immune-tumor interactions and for designing neural or psychological interventions in cancer.

Strengths of the Study

- **Multi-modal design**
 - Integrates sc/snRNA-seq, spatial transcriptomics, mIHC, and *in vitro* functional assays.
 - Enables cross-validation of cell states and spatial patterns around neural invasion (NI).
- **Focus on neural invasion microenvironment**
 - Explicitly defines NI-high vs NI-low tumors and juxta-neural regions.
 - Dissects malignant, Schwann, myeloid, and fibroblast subtypes in NI context rather than globally.
- **Comprehensive cell-state annotation**
 - Multiple malignant ductal subtypes, distinct Schwann clusters (including TGFBI⁺), diverse macrophage and CAF subsets.
 - Combines CNV inference, gene signatures, and pathway scores for annotation.

Strengths of the Study

- **Spatially resolved analyses**
 - Uses spatial transcriptomics and NI-focused regions to map gradients of cell states around invaded nerves.
 - Correlates Schwann and myeloid subsets with NI severity and nerve distance.
- **Network and communication analysis**
 - Applies tools such as network modeling (e.g., MistyR) and ligand-receptor analysis (e.g., CellChat/others).
 - Identifies candidate signaling axes between malignant cells, Schwann cells, macrophages, and CAFs.
- **Functional validation**
 - *In vitro* experiments support a role of TGF- β in inducing TGFBI⁺ Schwann phenotype.
 - Conditioned media and co-culture assays show enhanced PDAC migration/invasion toward TGF- β -stimulated Schwann cells.

Limitations and Potential Biases

- **Cohort size and representativeness**
 - Moderate patient numbers; single-center, mainly one population.
 - May limit generalizability across ethnicities, treatment backgrounds, or earlier disease stages.
- **Cross-sectional design**
 - All samples taken at surgery; no longitudinal sampling.
 - Hard to distinguish drivers vs consequences of NI (correlation rather than causation).
- **Definition and scoring of NI**
 - Histological NI scoring and region selection (juxta-neural) may involve pathologist subjectivity.
 - Cutoffs for NI-high vs NI-low may not directly translate to other centers.
- **Clinical annotation**
 - Limited information on prior treatments, comorbidities, and detailed outcomes.
 - NI-associated cell states not fully tested as independent prognostic factors in large cohorts.

Methodological Considerations

- **Single-cell vs single-nucleus and FFPE vs fresh tissue**
 - Different platforms and tissue handling introduce batch effects and cell-type coverage biases.
 - Some fragile or large cells (e.g., acinar, certain immune subsets) may be under-represented.
- **Malignant vs non-malignant assignment**
 - Relies on inferred CNV profiles and marker genes.
 - Tumor cells with low CNV or stromal cells with genomic alterations may be misclassified.
- **Spatial resolution limits**
 - Spot-based spatial transcriptomics cannot resolve single cells.
 - Deconvolution depends on sc/snRNA-derived signatures; errors propagate to NI-gradient analysis.
- **Pathway and network inference**
 - Gene-set enrichment and ligand-receptor predictions are model-based and sensitive to databases.
 - Network tools (e.g., MistyR) show associations, but do not prove direct physical interactions.

Clinical Relevance of NI-Associated Cell States

- **Potential prognostic biomarkers**

- NI-associated malignant ductal subtypes (e.g., specific GABRP⁺ / CEACAM⁺ states).
- TGFBI⁺ Schwann subset enriched near invaded nerves.
- NI-linked macrophage and CAF states (e.g., NLRP3⁺ macrophages, MMP11⁺ myCAFs).

- **Patient stratification**

- Combining NI histology with molecular cell-state signatures may refine risk prediction.
- Could help identify patients at higher risk of local recurrence or neuropathic pain.

- **Companion diagnostics**

- mIHC / spatial markers around nerves may be developed into pathology panels.
- Liquid biopsy potential if NI-related signatures can be detected in blood or CSF (hypothesis-generating).

Therapeutic Opportunities (Conceptual)

- **Targeting TGF- β –Schwann axis**
 - TGF- β induces TGFBI⁺ Schwann phenotype that promotes tumor migration/invasion.
 - TGF- β pathway inhibitors (e.g., TGF- β R blockers) could be explored to limit pro-NI Schwann reprogramming.
- **Reprogramming NI-associated myeloid cells**
 - NLRP3⁺ or other pro-inflammatory macrophage subsets accumulate near nerves.
 - Targeting inflammasome, cytokines, or macrophage polarization may modulate the NI niche.
- **Modulating CAF and ECM components**
 - MyCAF subsets and matrix-remodeling genes enriched near invaded nerves.
 - Local stromal modulation (e.g., ECM-targeting, CAF reprogramming) might reduce nerve tropism.
- **Combination strategies**
 - Integrate NI-focused interventions with chemotherapy, radiotherapy, or immunotherapy.
 - Rational combinations require careful toxicity and off-tumor effects assessment (nerves, normal Schwann cells).

Thanks!